

SPRING BOOT – FIRST LAB

Book API – Lab Submission Template

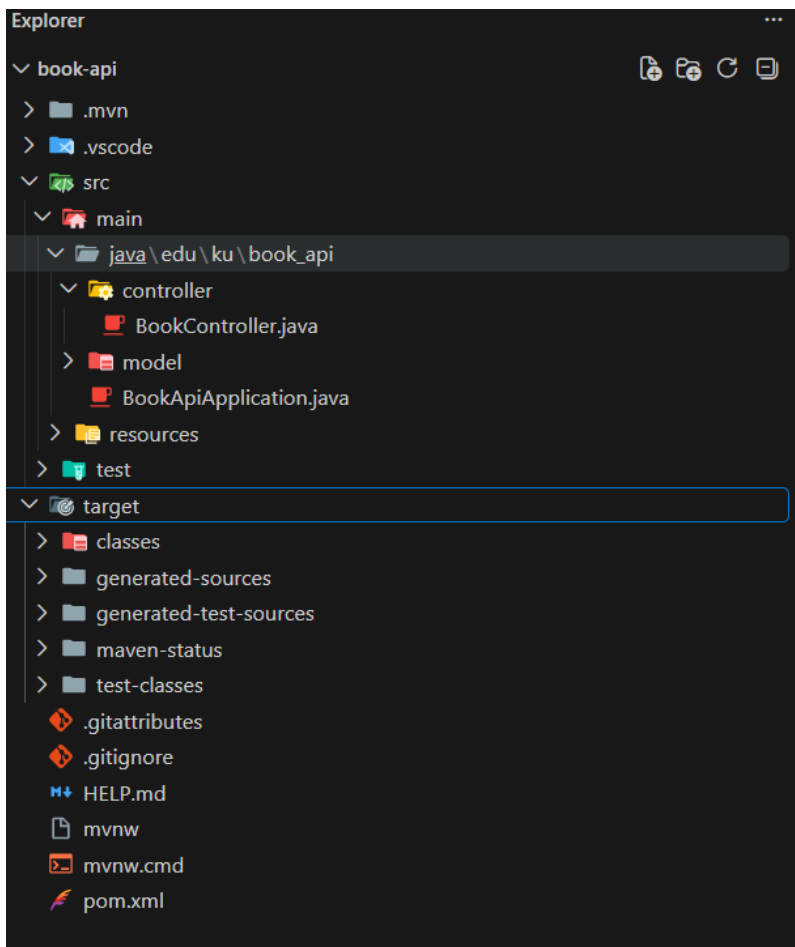
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Semester	7th Semester
Assignment	First Lab – Book API

1. Project Overview

This project is a basic Spring Boot REST API for managing and returning a list of books. The application contains a model class named Books and a REST controller named BookController. The main endpoint is GET /api/v1/books.

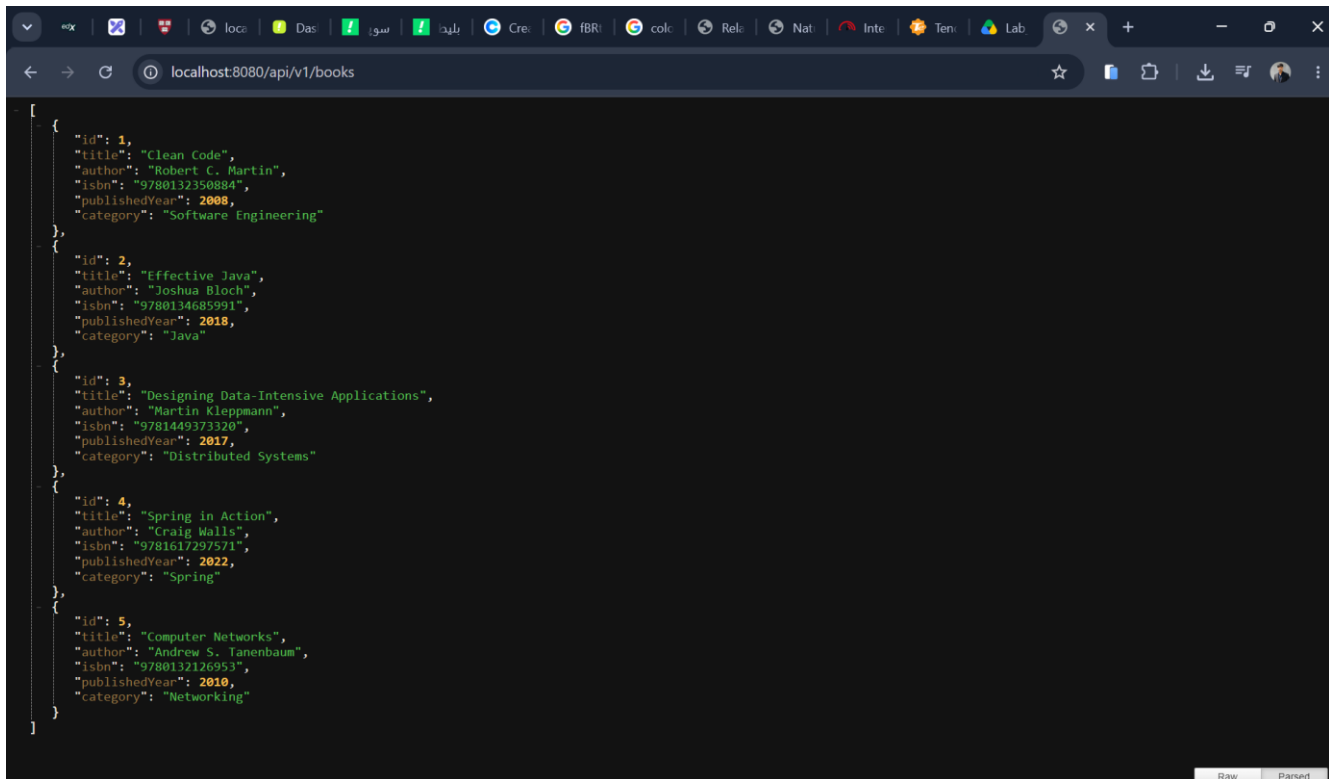
2. Project Folder Structure

The project should follow a structure similar to the following. Adjust the package name if your actual project is different.



3. Required Screenshot – GET /api/v1/books

Paste a screenshot from Postman, browser, or curl showing GET /api/v1/books, HTTP 200, and the returned JSON list.



```
[{"id": 1, "title": "Clean Code", "author": "Robert C. Martin", "isbn": "9780132350884", "publishedYear": 2008, "category": "Software Engineering"}, {"id": 2, "title": "Effective Java", "author": "Joshua Bloch", "isbn": "9780134685991", "publishedYear": 2018, "category": "Java"}, {"id": 3, "title": "Designing Data-Intensive Applications", "author": "Martin Kleppmann", "isbn": "9781449373320", "publishedYear": 2017, "category": "Distributed Systems"}, {"id": 4, "title": "Spring in Action", "author": "Craig Walls", "isbn": "9781617297571", "publishedYear": 2022, "category": "Spring"}, {"id": 5, "title": "Computer Networks", "author": "Andrew S. Tanenbaum", "isbn": "9780132126953", "publishedYear": 2010, "category": "Networking"}]
```

4. Reflection Questions and Answers

1. Why is GET appropriate for /api/v1/books?

GET is appropriate because the endpoint retrieves book data from the server. GET is the standard HTTP method for reading resources and normally does not modify server data. Therefore, GET /api/v1/books requests the collection of books.

2. What is the purpose of @RestController?

@RestController tells Spring that the class is a REST controller. It combines @Controller and @ResponseBody, allowing methods to handle HTTP requests and return data directly in the response body, commonly as JSON.

3. What does <Books> mean in List<Books>?

<Books> is the generic type of the Java List. It means the list contains objects of the Books class. Thus, List<Books> represents a collection of Books objects.

4. Why is books plural in the URI, and what does v1 represent?

books is plural because the URI represents a collection of book resources rather than one specific book. v1 means version 1 of the API. Versioning allows future API versions to introduce changes while keeping the existing version available.

5. How does a Java List<Books> become JSON without manually writing JSON code?

Spring Boot uses HTTP message converters, commonly with the Jackson JSON library, to serialize Java objects into JSON automatically. When the controller returns List<Books>, Spring converts the list and its Books objects into JSON and places it in the HTTP response.

6. What would happen if BookController were placed outside the package hierarchy scanned by the Spring Boot application?

Spring Boot would normally not discover the BookController during component scanning. The controller would not be registered as a Spring bean, so /api/v1/books would not be available and a request would typically return 404 Not Found. The solution is to place the controller under the application's scanned package or explicitly configure component scanning.

Conclusion

This lab demonstrates the fundamental structure of a Spring Boot REST API. The Books model represents book data, while BookController exposes the /api/v1/books endpoint. The application was started successfully and the endpoint was tested using an HTTP client or browser. The reflection questions explain the main Spring Boot, REST, Java generics, API versioning, JSON serialization, and component-scanning concepts used in the lab.